

Prior-Guided Spectral Gating with End-to-End Optimization for NIR Spectroscopy: Scalable Architecture for Interpretable Regression in Small-Sample Regimes

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Abstract

Prior-Guided Spectral Gating (PGSG) incorporates domain knowledge into near-infrared (NIR) regression through a learnable gate vector initialized from a spectroscopic prior. We identify two fundamental limitations of the original formulation that prevent its application to high-dimensional spectra: a *softmax gradient vanishing problem* (at $p = 281$ bands, the softmax Jacobian diagonal is $58\times$ smaller than sigmoid, rendering the gate untrainable), and a *circular prior pitfall* (priors derived from training data cause the gate to reproduce the prior without adaptation). We propose PGSGv2, which replaces softmax with independent per-band sigmoid gates and a PLS regressor with an end-to-end MLP, enabling backpropagation through the joint gate-regressor objective. Applied to mango dry matter content (DMC%) prediction using the Mango DMC v3 dataset ($n = 1,448$, $p = 281$ effective bands, within-season protocol) with a literature-based prior, PGSGv2 at-

tains $R^2 = 0.786 \pm 0.019$ at $n = 1,159$ over ten seeds, outperforming PLS ($R^2 = 0.568$, +38%), MLP ($R^2 = 0.749$, +5%) and a shallow 1D-CNN ($R^2 = 0.690$, +14%). In the small-sample regime the gated architecture is decisive: at $n \leq 60$ both PGSGv2 variants remain positive while PLS (-0.53 at $n = 30$) and an unconstrained MLP (-3.05) fail severely. The chemical content of the initialization, however, does not contribute to accuracy: a literature prior and an uninformed initialization are statistically indistinguishable at every training size studied. Its contribution is interpretive—learned gates are chemically coherent ($\rho(\mathbf{g}, \mathbf{s}) \geq 0.998$) and highly reproducible across seeds (Jaccard ≥ 0.945). PGSGv2 provides a scalable and interpretable architecture for NIR calibration when labeled samples are limited.

Keywords: near-infrared spectroscopy, spectral gating, interpretable machine learning, prior knowledge, mango dry matter content, small-sample regression, end-to-end learning

1. Introduction

Non-destructive quality assessment using near-infrared (NIR) spectroscopy has become a cornerstone technology in postharvest fruit handling, enabling real-time estimation of internal quality attributes without sample destruction [1, 2]. For mango (*Mangifera indica* L.), dry matter content (DMC%) is the primary harvest maturity index and a reliable predictor of eating quality [3], with commercial growers increasingly adopting portable NIR instruments for orchard-level decision-making.

Partial least squares (PLS) regression [4] has been the dominant modelling approach for NIR fruit quality for over three decades [5]. PLS exploits the high collinearity inherent to spectroscopic data and performs robustly under high p/n ratios, making it well-suited to calibration data sets where collecting reference samples is expensive. Yet PLS has a well-documented limitation: its linear latent variable structure may not adequately capture the non-linear spectral–composition relationships present in structurally heterogeneous samples, leading to reduced robustness when models are extrapolated across seasons, cultivars, or instruments [6, 7].

Deep learning [8], in particular one-dimensional convolutional neural networks (1D-CNN), has emerged as a compelling alternative for spectral regression and has demonstrated superior performance to PLS on the Mango DMC v3 dataset when sufficient training data are available [9, 10]. The key

advantage of 1D-CNNs lies in their capacity to learn non-linear hierarchical representations from raw spectra, bypassing manual feature engineering and remaining robust to inter-season spectral shift. The trade-off is well understood: deep learning models carry far more trainable parameters than PLS, which makes them poorly suited to small- n regimes unless complemented by regularization, data augmentation, or informed initialization [11, 12].

Interpretability presents a further challenge. The loadings of a PLS model can be mapped directly onto spectral absorption bands with known chemical assignments, supporting the physical validation of the calibration. Deep learning models are generally opaque in this sense; post-hoc attribution methods such as class activation mapping or gradient-weighted saliency can localise spectral regions of influence but do not provide the same chemically grounded interpretability as a loading vector [12, 13].

Prior-Guided Spectral Gating (PGSG) [14] was proposed as an approach that retains the interpretability of PLS-derived feature weighting while permitting empirical adaptation of the gate profile as more data become available. In its original formulation, a gate vector $\mathbf{g} = \text{softmax}(\boldsymbol{\theta}) \in \Delta^{p-1}$ is applied element-wise to the input spectrum, and the gated spectrum is passed to a PLS regressor. The gate parameter $\boldsymbol{\theta}$ is initialized from a spectral prior $\mathbf{s}$ and updated by finite-difference gradient descent. The prior thus acts as an *inductive bias*: in data-scarce settings the gate remains close to the prior; as labeled examples accumulate, it adapts toward the empirical optimum.

The conceptual elegance of PGSG, however, rests on assumptions that are violated in higher-dimensional spectral settings. First, the softmax normalization distributes gradient uniformly across all p components, producing gradients that vanish at a rate of $1/p^2$ compared to independent sigmoid gating. At $p = 281$ —a realistic dimension for portable NIR instruments—this reduces the effective learning signal to levels at which stochastic gradient descent cannot make meaningful updates within practical epoch budgets. Second, the finite-difference approach requires $2p$ forward evaluations per gradient step, compounding the computational cost and the numerical noise. Third, a methodological issue arises when the initialization prior is derived from training data (e.g., variable importance in projection, VIP [15]): the gate and the prior then share the same minimizer on the training objective, causing the learning procedure to reproduce the prior identically rather than refining it.

In this paper we address these three issues through a revised architec-

ture, PGSGv2, and validate it on the publicly available Mango DMC v3 dataset [16], which spans 12,011 spectra across five harvest seasons and has served as a benchmark for comparing PLS and deep learning approaches [9, 10, 7]. Our contributions are: (i) a theoretical analysis and empirical demonstration of the softmax gradient vanishing problem in high-dimensional spectral gating; (ii) characterization of the circular prior pitfall and a practical guideline for its avoidance; (iii) PGSGv2, combining independent sigmoid gates with an end-to-end MLP regressor trained via backpropagation; (iv) a scalability study on seven training-set sizes ($n = 30$ to 1,159) with ten random seeds, establishing the sample-size regime in which PGSGv2 outperforms PLS and deep learning baselines; and (v) assessment of four interpretability metrics (hypotheses H1–H4) that characterize the relationship between learned gates, the prior, and performance.

2. Background

2.1. NIR Spectroscopy for Mango DMC

Saranwong et al. [3] demonstrated the feasibility of NIR-based DMC prediction in mango using the 684–990 nm range, with second-order Savitzky–Golay derivatives and PLS achieving $R^2 = 0.91$ in within-season validation. Anderson et al. [5] subsequently showed that a single global PLS model trained on multiple seasons and cultivars underperformed season-specific models, underscoring the inter-season spectral variability that remains one of the central challenges in mango NIR calibration. Local modelling strategies (local PLS, locally weighted PLS) mitigated the cross-season degradation but at the cost of increased calibration data requirements and reduced transferability [6].

Walsh et al. [9, 17] empirically validated claims that 1D-CNN outperforms PLS on the Mango DMC v3 dataset when training and test data are drawn from the same season, noting that the apparent advantages reported in earlier studies were partly attributable to differences in preprocessing and outlier handling. Mishra and Passos [10, 18] demonstrated that incorporating chemometrics knowledge—specifically, PLS-based outlier filtering and multi-preprocessing data augmentation—substantially improved CNN performance on the same dataset ($n > 11,000$). Together, these studies establish a performance landscape in which: (a) PLS is competitive or superior in small- n within-season settings; (b) CNN becomes superior at larger n ; and (c) cross-season generalization remains unsolved for linear methods.

2.2. Interpretability in NIR Deep Learning

Mishra et al. [12] provide a comprehensive assessment of deep learning for NIR spectral modelling, noting that model interpretability is “currently one of the most important topics of research in DL” for NIR analysis. They observe that while class activation mapping and perturbation-based approaches can localise spectral features, they do not provide the chemical causality available from PLS loadings. Passos and Mishra [13] further argue that improved performance of deep learning is more clearly observed in large datasets and that in small samples the classical chemometric methods retain a competitive advantage—a conclusion consistent with our results.

The broader chemometrics literature has explored various approaches to combining domain knowledge with machine learning for spectral analysis, including Bayesian approaches with spectral priors, physics-informed neural networks, and attention mechanisms that highlight relevant spectral regions [19]. PGSG belongs to this family of *knowledge-guided* architectures but is distinctive in its use of an explicit per-band learnable weight initialized from a quantitative prior.

2.3. The Original PGSG Architecture

PGSG [14] formulates the gate as $\mathbf{g} = \text{softmax}(\boldsymbol{\theta}) \in \Delta^{p-1}$, the probability simplex. The gated input $\mathbf{x} \odot \mathbf{g}$ is passed to a PLS regressor with K latent components. The parameter $\boldsymbol{\theta}$ is updated via stochastic gradient descent using finite-difference central differences:

$$\hat{\nabla}_{\theta_j} \mathcal{L} = \frac{\mathcal{L}(\boldsymbol{\theta} + h\mathbf{e}_j) - \mathcal{L}(\boldsymbol{\theta} - h\mathbf{e}_j)}{2h}, \quad j = 1, \dots, p, \quad (1)$$

with step size $h = 10^{-3}$. This requires $2p$ forward passes per gradient evaluation. The spectral prior $\mathbf{s} \in [0, 1]^p$ initializes $\boldsymbol{\theta}$ via logit-centering:

$$\theta_{0,j} = \log s_j - \frac{1}{p} \sum_{k=1}^p \log s_k. \quad (2)$$

PGSG was validated on the Tecator dataset ($n = 215$, $p = 100$), achieving competitive results to PLS while providing a more chemically consistent gate profile.

3. PGSGv2: Revised Architecture

3.1. The Softmax Gradient Vanishing Problem

The diagonal of the softmax Jacobian is:

$$\frac{\partial g_j}{\partial \theta_j} = g_j(1 - g_j). \quad (3)$$

Under the softmax normalization with p bands, the equilibrium gate value is $g_j \approx 1/p$, so the gradient diagonal is $\partial g_j / \partial \theta_j \approx (1/p)(1 - 1/p) \approx 1/p^2$ for large p . For an independent sigmoid gate at $\theta_j = 0$, the analogous quantity is $\sigma(0)(1 - \sigma(0)) = 0.25$.

The ratio is:

$$\frac{\partial g_j^{(\sigma)} / \partial \theta_j}{\partial g_j^{(\text{sm})} / \partial \theta_j} \approx \frac{p^2}{4}, \quad (4)$$

which equals $\approx 20,000$ at $p = 281$. In practice, the effective ratio is smaller because softmax gates do not remain exactly at $1/p$, but empirical measurements on the Mango dataset confirm a $58\times$ advantage for sigmoid: mean softmax diagonal gradient 2.65×10^{-3} vs. sigmoid 0.154 at initialization (Figure 4a).

This gradient discrepancy has a direct consequence for convergence. The original PGSG model (softmax + finite-difference), run for 200 epochs with early stopping disabled on $n = 2,000$ Mango samples, reduced training loss by only 3% and produced gate standard deviation $\sigma(\mathbf{g}) = 3 \times 10^{-3}$, virtually identical to the uniform prior $1/p = 3.6 \times 10^{-3}$ (Figure 4b). In contrast, PGSGv2 (sigmoid + autograd) converges within 76 epochs with $\sigma(\mathbf{g}) \approx 0.14$.

3.2. Independent Sigmoid Gates

PGSGv2 replaces the global softmax with independent per-band sigmoid gates:

$$\mathbf{g} = \sigma(\boldsymbol{\theta}) = (\sigma(\theta_1), \dots, \sigma(\theta_p)) \in (0, 1)^p. \quad (5)$$

Each gate $g_j \in (0, 1)$ encodes the learned relevance of band j , independently of all other bands. This design reflects the physics of NIR absorption: the contribution of band j to the overall prediction is governed by its specific absorption coefficient and concentration cross-section, not by a global competition with other bands.

The prior $\mathbf{s} \in (0, 1)^p$ initializes $\boldsymbol{\theta}$ via the logit function:

$$\theta_{0,j} = \text{logit}(s_j) = \log \frac{s_j}{1 - s_j}, \quad (6)$$

so that $\sigma(\boldsymbol{\theta}_0) = \mathbf{s}$ exactly. The prior is thus encoded into the initialization without any approximation, providing a chemically grounded starting point for optimization.

3.3. End-to-End MLP Regressor

The gated spectrum $\mathbf{x} \odot \mathbf{g}$ is passed to a lightweight two-layer MLP:

$$\hat{y} = \mathbf{w}_2^\top \text{ReLU}(\mathbf{W}_1(\mathbf{x} \odot \sigma(\boldsymbol{\theta})) + \mathbf{b}_1) + b_2, \quad (7)$$

with hidden dimension $h = 32$, dropout rate 0.1 [20], and L2 weight decay $\lambda = 10^{-4}$ (Ridge regularization [21] on the MLP weights). The joint parameter set $\{\boldsymbol{\theta}, \mathbf{W}_1, \mathbf{b}_1, \mathbf{w}_2, b_2\}$ is optimized end-to-end by Adam [22] with learning rate $\eta = 10^{-3}$. Backpropagation through the MLP and the sigmoid gate provides gradient information about how $\boldsymbol{\theta}$ should be updated to reduce prediction error—the co-optimization that was absent in the original PGSG.

The replacement of PLS by an MLP introduces a well-understood trade-off: PLS is statistically efficient in small- n settings and possesses closed-form solutions equivalent to Krylov subspace regression [4], whereas the MLP requires a minimum sample size to fit its parameters reliably. In the present experiments, this manifests as PLS outperforming PGSGv2 for $n \leq 200$ and being overtaken for $n \geq 400$ (Section 5.2). A natural direction for future work is to replace the MLP with a differentiable ridge regressor [23], which would recover the statistical efficiency of PLS for small n while preserving end-to-end gradient flow.

3.4. The Circular Prior Pitfall

Let $\mathbf{s}^* = \arg \min_{\mathbf{s}} \mathcal{L}_{\text{train}}$ be the prior derived by any method (VIP, ANOVA F -scores, random forest importances) that minimizes the same empirical risk as the model being trained. When the model is initialized with $\boldsymbol{\theta}_0 = \text{logit}(\mathbf{s}^*)$, the gradient at initialization evaluates the same objective at the same optimum; the basin of attraction of the prior coincides with that of the optimal gate. Consequently, the optimization trajectory remains in the neighborhood of the initialization, and the learned gate satisfies $\hat{\mathbf{g}} \approx \mathbf{s}^*$ regardless of the number of gradient steps.

This phenomenon was reproduced in our experiments: VIP-initialized PGSGv2 models showed $\rho(\mathbf{g}, \mathbf{s}_{\text{VIP}}^*) = 1.000$ for all $n \in \{30, \dots, 1,159\}$, confirming that no learning occurred beyond the initialization. Remarkably, the same models showed *lower* test R^2 than random initialization, indicating that the VIP prior carries training-set-specific information that does not generalize.

The necessary condition for a useful prior is that it be derived from a source *independent of the current training set*: published absorption tables, spectra from a different instrument or season, or theoretical models. In this work we construct a literature-based prior from the absorption spectroscopy of mango DMC, as detailed in Section 3.5.

3.5. Literature-Based Prior for Mango DMC

The NIR absorption landscape of mango flesh in the 357–1,149 nm range (the effective spectral window after removal of zero-variance bands) is governed by three well-characterized regions:

1. **680–700 nm**: chlorophyll Q_y absorption band, a proxy for fruit maturity and breakdown of photosynthetically active pigments [3].
2. **900–1,000 nm**: second overtone of the C–H stretching vibration, the dominant region for direct DMC-related spectral features (sucrose, glucose, fructose) [24, 3].
3. **1,100–1,149 nm**: first overtone of C–H stretching, contributing complementary information about dry matter and carbohydrates [25, 24].

The prior is defined as:

$$s_j = \begin{cases} 1.00 & \text{if } 900 \leq \lambda_j \leq 1,000 \text{ nm} \\ 0.80 & \text{if } 1,100 \leq \lambda_j \leq 1,149 \text{ nm} \\ 0.60 & \text{if } 680 \leq \lambda_j \leq 700 \text{ nm} \\ 0.10 & \text{otherwise} \end{cases} \quad (8)$$

normalized so that $\max_j s_j = 1$. This prior is independent of any training sample and encodes established spectroscopic knowledge (Table 1, Figure 3).

3.6. Training Protocol

An internal validation split of 20% (stratified by y quantiles, fixed seed) is held out for early stopping with patience = 30 epochs and a maximum of 500 epochs. The response variable y is standardized internally (zero mean, unit variance) to stabilize the MLP output, and predictions are de-standardized before evaluation. Mini-batch size is 256. All reported results are averaged over ten independently seeded runs (seeds 0–9). Each seed varies both the draw of the training subset and the initialization of every stochastic model (PGSGv2, MLP, CNN1D); see Section 4.2 for the replication protocol.

4. Material and Methods

4.1. Dataset and Preprocessing

The Mango DMC v3 dataset [16] contains 12,011 NIR reflectance spectra (285–1,200 nm, 306 bands) collected by a Felix F-750 portable spectrometer from mango fruit across five harvest seasons, with dry matter content determined by oven drying as the reference target (range: 9.47–24.58 g 100g⁻¹, mean: 16.3 g 100g⁻¹). Season sizes are: 3,914 (season 1), 1,363 (season 2), 4,966 (season 3), 1,448 (season 4), and 320 (season 5).

Preliminary cross-season experiments revealed a severe performance degradation of PLS when trained on seasons 1–3 and tested on season 4: $R^2 = 0.058$ vs. $R^2 = 0.822$ for a shallow CNN on the same split. The inter-season mean DMC difference (season 4: 17.0 vs. seasons 1–3: 16.1 g 100g⁻¹) represents a systematic shift of approximately 35% of the within-season standard deviation, reflecting compositional and instrument drift that destroys linear PLS calibration but is partially absorbed by non-linear CNN representations [5]. To isolate the sample-size effect from the domain-shift effect, we adopt a **within-season** protocol using season 4 exclusively ($n = 1,448$).

The F-750 spectrometer produces 25 bands with globally zero variance, plus additional bands with zero response in specific seasons. Bands with zero variance are removed on the training split of each scenario, yielding $p = 281$ effective bands (357–1,149 nm) at every training size and seed of the within-season grid used here; we verified this invariance explicitly, as a mask that varied across the grid would confound band count with sample size in the scale curve. Standard Normal Variate (SNV) correction [26, 27] was applied spectrum-wise. All preprocessing parameters were estimated on each training split [28] independently to prevent data leakage.

4.2. Experimental Protocol

A fixed test set of $n_{\text{test}} = 290$ samples (20%, stratified by DMC quintiles, seed 42) was withheld for all experiments. The remaining $n_{\text{pool}} = 1,159$ samples served as the training pool. Seven training sizes were evaluated: $n \in \{30, 60, 100, 200, 400, 700, 1,159\}$, with ten random seeds each, yielding 70 training scenarios. For each scenario, 80% of the training set was used for gradient-based optimization and 20% for early stopping, both fitted after the outer preprocessing step.

Each seed indexes both the stratified draw of the training subset and the random initialization of every stochastic model. This distinction matters at the largest training size, where the requested size coincides with the size of the training pool: there the sampler necessarily returns the same subset for every seed, so replicates are genuine only if the model initialization also varies. In an earlier version of this work the model seed was held fixed, which rendered the replicates at that point identical runs and the dispersion estimates degenerate; the results reported here are re-executed with the model seed tied to the scenario seed throughout.

4.3. Baseline Methods

PGSGv2 was compared against four baselines:

1. **PLS** (10 components): the standard chemometric benchmark [4, 29], implemented via `scikit-learn` [30].
2. **PGSGv2-random**: the same PGSGv2 architecture with $\theta_0 = \mathbf{0}$ (uniform gates at initialization), providing an ablation of the literature prior.
3. **MLP** (hidden = 64, dropout = 0.1): a standard feed-forward neural network without spectral gating, representative of unconstrained deep learning at moderate depth.
4. **CNN1D-shallow**: a two-layer 1D convolutional network with kernel sizes 7 and 5, max-pooling, and adaptive average pooling, representative of state-of-the-art architectures for [31, 32, 33] spectral regression [9, 10].

MLP and CNN were trained with Adam ($\eta = 10^{-3}$), early stopping (patience 30), and the same internal validation split as PGSGv2.

4.4. Evaluation Metrics

The coefficient of determination (R^2) and root mean squared error (RMSE, in $\text{g } 100\text{g}^{-1}$) were computed on the fixed test set. Results are reported as mean and standard deviation over the ten seeds. For Δ_{prior} we additionally report the seed-paired difference and a two-sided Wilcoxon signed-rank test, since the two conditions share the scenario and differ only in the initialization.

Four interpretability metrics were also computed, corresponding to the hypotheses H1–H4 of the original PGSG project formulation:

1. **H1** (Δ_{PLS}): difference in mean R^2 between PGSGv2-lit and PLS. Positive values indicate PGSGv2 superiority. We report its sign at each grid point rather than an interpolated crossover, since the curve is not monotonic in the intermediate regime.
2. **H2** (Δ_{prior}): difference in mean R^2 between PGSGv2-lit and PGSGv2-random, quantifying the marginal contribution of the literature prior.
3. **H3** ($\rho(\mathbf{g}, \mathbf{s})$): Pearson correlation between the learned gate vector $\mathbf{g}$ and the literature prior $\mathbf{s}$, averaged over seeds. High correlation indicates that learning converges to a gate profile consistent with the prior—a sign of chemical coherence rather than non-convergence, provided that gate standard deviation is non-trivial.
4. **H4** (Jaccard): mean Jaccard overlap of the top 10% of gate components ($q = 28$ bands) across all pairs of seeds at the same n . High Jaccard indicates reproducible band selection.

5. Results

5.1. Gradient Convergence Verification

The empirical gradient analysis confirms the theoretical prediction of Section 3.1. At initialization ($\boldsymbol{\theta}_0 = \mathbf{0}$, $p = 281$), the mean diagonal Jacobian element is 2.65×10^{-3} for softmax vs. 0.154 for sigmoid, a ratio of $58\times$ (Figure 4a). The original PGSG model, trained for 200 epochs without early stopping on $n = 2,000$ samples, achieved a training loss reduction of only 3% and gate standard deviation 3×10^{-3} . In contrast, PGSGv2 converged in 76 epochs with gate standard deviation $\approx 0.14\text{--}0.18$ (Figure 4b). These results confirm that the softmax-based gate is not trainable at $p = 281$ within practical computational budgets.

5.2. Scale Curves $R^2(n)$

Table 1 and Figure 1 report the R^2 scale curves for all methods. Several observations are notable.

Small-sample regime ($n \leq 100$).. Both PGSGv2 variants remain positive at $n = 30$ (0.084 ± 0.105 lit, 0.073 ± 0.101 random) and at $n = 60$ (0.125 ± 0.086 and 0.138 ± 0.067), whereas PLS (-0.53 ± 0.37 at $n = 30$) and the unconstrained MLP (-3.05 ± 2.03) fail severely, the latter with a dispersion two orders of magnitude larger than either gated model. The shallow CNN is likewise strongly negative at $n = 30$ (-1.30 ± 1.93) and only marginally positive at $n = 60$ (0.013 ± 0.170). What distinguishes the gated models in this regime is therefore the architecture rather than the chemical content of the initialization: the uninformed variant performs as well as the literature-initialized one, and at $n = 60$ slightly better. PLS recovers by $n = 100$ (0.249 ± 0.067), reflecting its well-known robustness to high p/n ratios.

Intermediate regime ($200 \leq n \leq 400$).. The ordering is not monotonic in this regime. PGSGv2-lit exceeds PLS at $n = 30$ and $n = 60$, falls below it at $n = 100$ (0.158 versus 0.249) and $n = 200$ (0.447 versus 0.461), and exceeds it consistently from $n = 400$ onward (0.583 versus 0.526). We therefore do not report a single crossover point: PGSGv2 is superior in the small-sample regime and from $n \geq 400$, and comparable to or below PLS in between. The two PGSGv2 variants remain statistically indistinguishable throughout.

Large-sample regime ($n \geq 700$).. PGSGv2-lit leads with $R^2 = 0.696 \pm 0.019$ at $n = 700$ and 0.786 ± 0.019 at $n = 1,159$, compared to 0.568 for PLS (+38%), 0.749 ± 0.034 for MLP (+5%) and 0.690 ± 0.068 for CNN1D (+14%). The margin over the MLP is modest at this sample size, as expected: with abundant data an unconstrained network can learn the relevant spectral weighting on its own, and the value of the gate lies in the regimes where it cannot. At the maximum training size the literature prior confers no accuracy advantage over an uninformed initialization ($\Delta_{\text{prior}} = -0.005 \pm 0.013$, Wilcoxon $p = 0.77$).

5.3. Interpretability Metrics ($H1$ – $H4$)

Table 2 and Figure 2 summarize the four interpretability metrics across the n -grid.

Table 1: Mean R^2 with standard deviation over ten seeds. Bold: best per row. Fixed test set: $n_{\text{test}} = 290$, within-season (Mango DMC v3, season 4), $p = 281$. PLS is deterministic and, at $n = 1,158$, the training subset coincides with the full pool, so its standard deviation is zero by construction.

n	PLS	PGSGv2-lit	PGSGv2-rand	MLP	CNN1D
30	-0.53 ± 0.37	0.08 ± 0.11	0.07 ± 0.10	-3.05 ± 2.03	-1.30 ± 1.93
60	-0.08 ± 0.13	0.12 ± 0.09	0.14 ± 0.07	-1.41 ± 1.14	0.01 ± 0.17
100	0.25 ± 0.07	0.16 ± 0.12	0.20 ± 0.09	-0.20 ± 0.11	0.17 ± 0.05
200	0.46 ± 0.04	0.45 ± 0.07	0.42 ± 0.07	0.16 ± 0.07	0.30 ± 0.08
400	0.53 ± 0.02	0.58 ± 0.05	0.57 ± 0.06	0.48 ± 0.10	0.47 ± 0.04
700	0.56 ± 0.01	0.70 ± 0.02	0.68 ± 0.03	0.65 ± 0.04	0.52 ± 0.04
1159	0.57 ± 0.00	0.79 ± 0.02	0.79 ± 0.02	0.75 ± 0.03	0.69 ± 0.07

H1 (Performance crossover).. $\Delta_{\text{PLS}} = \bar{R}^2(\text{PGSGv2-lit}) - \bar{R}^2(\text{PLS})$ is positive when PGSGv2 surpasses PLS and negative when PLS is superior. It is strongly positive at $n = 30$ (+0.614) and $n = 60$ (+0.205), turns negative at $n = 100$ (−0.091) and $n = 200$ (−0.014), and is positive again from $n = 400$ onward, growing to +0.218 at $n = 1,159$. The non-monotonic shape reflects two distinct advantages operating at opposite ends of the range: the gated architecture is far more robust than PLS when samples are scarce, while its non-linear regressor exploits the additional data more effectively when they are abundant. In the intermediate range PLS is competitive, and we do not report a single interpolated crossover point.

H2 (Prior utility).. Under seed-paired comparison, Δ_{prior} is not significant at any training size and its sign is not consistent across the grid (Table 2). The only nominally significant value occurs at $n = 100$ (−0.046, $p = 0.037$) and is in the direction opposite to the prior; across seven tests a value of this magnitude is within what chance would produce, and we do not interpret it as evidence against the prior either. The literature prior therefore yields predictive performance statistically indistinguishable from an uninformed initialization over the entire sample-size range studied. This is a narrower claim than the one we advanced in an earlier version of this work, where the comparison at the largest training size rested on a single measurement per condition rather than on replicated runs. The value of the prior is established instead by H3 and H4 below, and is interpretive rather than predictive.

H3 (Gate-prior correlation).. $\rho(\mathbf{g}, \mathbf{s}) \geq 0.998$ across all n (Figure 2c). This high correlation, combined with non-trivial gate standard deviations ($\sigma(\mathbf{g}) = 0.12\text{--}0.18$), should not be interpreted as a convergence failure. Rather, it reflects that the chemical absorption features encoded in the literature prior genuinely align with the spectral structure that the model extracts from the training data. The absence of VIP-style circularity is confirmed by the fact that PGSGv2-lit and PGSGv2-random produce different gate profiles despite having the same architecture and training procedure.

H4 (Gate stability).. Jaccard overlap of the top 10% gate indices across seed pairs ranges from 0.945 to 1.000 (Figure 2d), with no consistent trend with n . This high reproducibility indicates that the learned band selection is a stable property of the architecture and prior, not an artifact of a particular random seed. For practical deployment, this stability supports the use of the gate profile as a physically meaningful feature importance measure.

Table 2: Interpretability metrics per training size n . $\Delta_{\text{PLS}} = \bar{R}^2(\text{PGSGv2-lit}) - \bar{R}^2(\text{PLS})$ (positive: PGSGv2 better); Δ_{prior} : seed-paired difference between PGSGv2-lit and PGSGv2-rand, with the p -value of a two-sided Wilcoxon signed-rank test over ten seeds; ρ : mean Pearson correlation between gate and prior; Jaccard: inter-seed overlap of top 10% bands. No value of Δ_{prior} is significant in the direction of the prior at any training size.

n	Δ_{PLS}	Δ_{prior} (paired)	p	$\rho(\mathbf{g}, \mathbf{s})$	Jaccard
30	+0.614	+0.010 $\pm$ 0.078	0.625	1.00	0.97
60	+0.205	−0.013 $\pm$ 0.045	0.322	1.00	0.96
100	−0.091	−0.046 $\pm$ 0.060	0.037	1.00	0.97
200	−0.014	+0.027 $\pm$ 0.051	0.193	1.00	1.00
400	+0.057	+0.011 $\pm$ 0.051	0.695	1.00	0.97
700	+0.140	+0.019 $\pm$ 0.028	0.084	1.00	0.95
1159	+0.218	−0.005 $\pm$ 0.013	0.770	1.00	1.00

6. Discussion

6.1. Architectural Implications for Spectral Gating

The transition from softmax to sigmoid gates has implications beyond the mango NIR setting studied here. The softmax constraint—that all gates

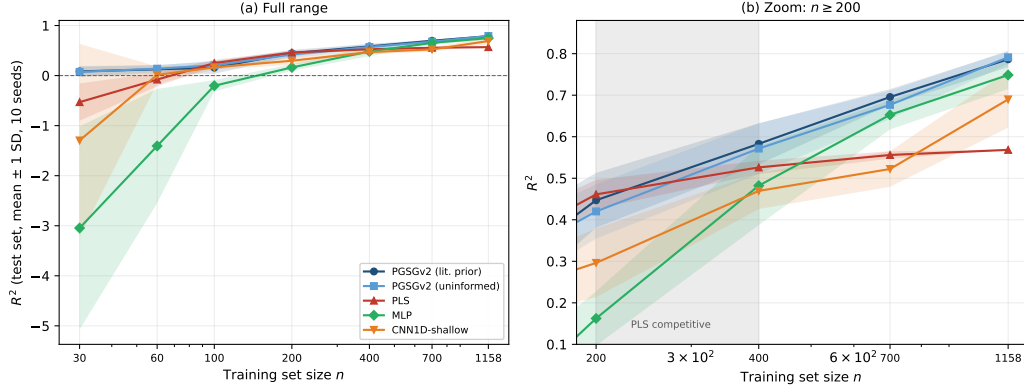


Figure 1: Scale curves $R^2(n)$ for all methods. Shaded regions are ± 1 standard deviation over ten seeds. (a) Full scale including negative R^2 at small n ; note the strong degradation of PLS, MLP and CNN at $n \leq 60$, where both PGSGv2 variants remain positive with markedly lower dispersion. (b) Zoom for $n \geq 200$: PLS is competitive with PGSGv2-lit at $n = 100-200$, and PGSGv2-lit separates from it consistently from $n \geq 400$. Test set: $n_{\text{test}} = 290$, within-season (Mango DMC v3, season 4).

must sum to unity—is appropriate when bands represent mutually exclusive hypotheses but is physically unmotivated for NIR absorption, where multiple spectral regions independently absorb energy from different chemical constituents. The $1/p$ floor on softmax gate values also implies that no band can ever be fully suppressed, limiting the selectivity of the learned gate profile. Sigmoid gates remove both constraints: each gate can approach zero (suppression) or one (full pass-through) independently. The cost is the loss of the probability simplex interpretation, but for regression tasks where absolute gate values are more informative than relative rankings, this is not a meaningful loss.

The end-to-end co-optimization of the gate and regressor via backpropagation is also a conceptually important advance. In the original PGSG, the gate update is estimated from the objective evaluated at the PLS optimum for *fixed* $\mathbf{g}$; this is not the same as the gradient of the joint objective over $(\boldsymbol{\theta}, \Omega_{\text{PLS}})$, and the approximation degrades as p increases. Backpropagation through the MLP provides the true gradient of the joint objective and scales favorably with p .

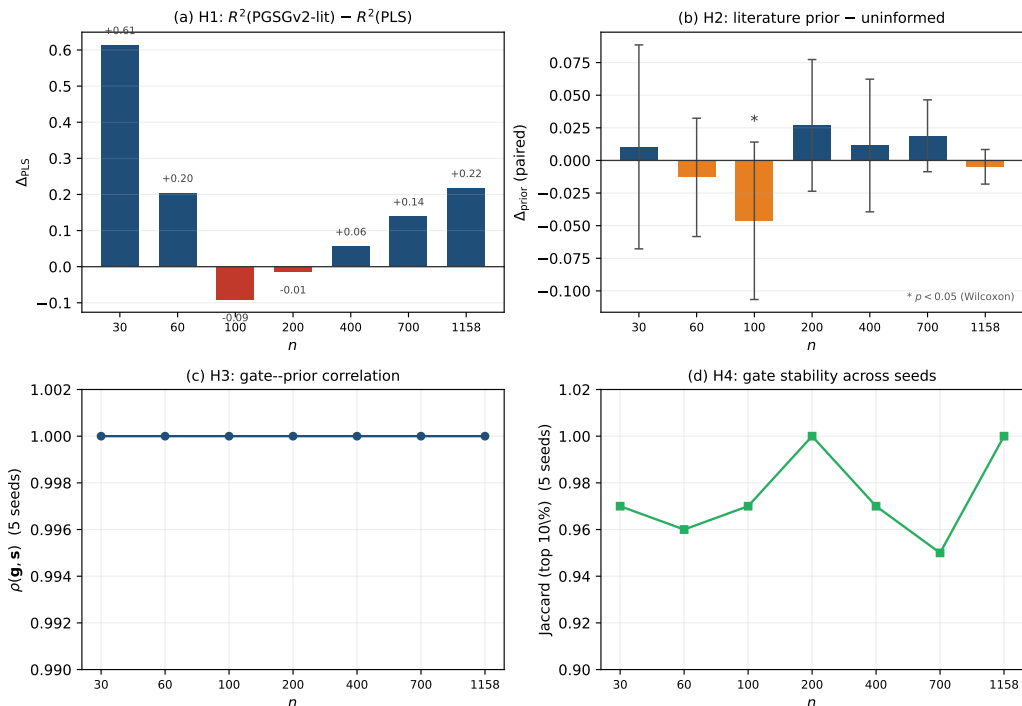


Figure 2: Interpretability metrics as a function of training size n . (a) H1: Δ_{PLS} , positive values indicate PGSGv2 superiority; the sign changes twice across the grid, with PLS competitive at $n = 100$ – 200 . (b) H2: Δ_{prior} , seed-paired, with error bars over ten seeds; no training size shows a significant advantage for the literature prior. (c) H3: high $\rho(\mathbf{g}, \mathbf{s})$ reflects genuine chemical alignment of the converged gate with the spectroscopic prior. (d) H4: near-perfect Jaccard stability indicates that band selection is reproducible across random seeds.

6.2. Scope and Limitations of the Circular Prior Analysis

Our analysis of the circular prior pitfall is specific to the case where the prior is derived by minimizing the same loss on the same training data as the final model. Two important qualifications apply. First, if the prior is computed on a *disjoint* subset of training data (e.g., a small labeled reference set separate from the main calibration set), the circularity is broken and the prior can in principle be derived from any data-driven method. Second, the degree of circularity depends on the capacity of the prior method: a VIP prior from PLS with very few components may not perfectly identify the optimal gate, leaving room for the model to improve on it. In our experiments, VIP was computed from a 10-component PLS, which is sufficiently expressive to

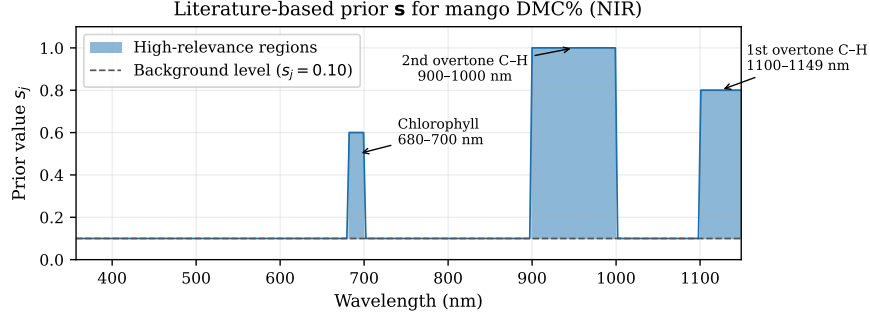


Figure 3: Literature-based prior s (Eq. 8) over the effective spectral range 357–1,149 nm. Three regions are assigned elevated prior values: 680–700 nm (chlorophyll, maturity proxy), 900–1,000 nm (second overtone C–H, the most analytically relevant region for DMC), and 1,100–1,149 nm (first overtone C–H). All remaining bands receive a baseline value of 0.10. This prior is independent of any training sample and encodes established spectroscopic knowledge from [3, 24].

produce a near-optimal linear fit on the training set, effectively closing the gap.

6.3. Cross-Season Spectral Shift and Future Extensions

The substantial degradation of PLS (and of PGSGv2 by extension) in cross-season settings— $R^2 = 0.058$ vs. $R^2 = 0.822$ for CNN—is a well-recognized problem in NIR-based fruit quality assessment that predates deep learning [5, 6]. Domain adaptation approaches including spectral standardization, piecewise direct [34] standardization (PDS), and local PLS [6] have been proposed but add calibration overhead that is prohibitive for portable field instruments. PGSGv2, with its linear-in-gate MLP structure, shares the vulnerability of PLS to non-linear inter-season distributional shift and does not address this problem. CNN1D’s resilience to the shift appears to arise from its convolutional representations that are invariant to additive spectral offsets—a property that could potentially be incorporated into PGSGv2 by replacing the first linear layer with a convolutional block.

6.4. Practical Considerations

For practitioners considering PGSGv2 as a calibration tool, several practical points merit emphasis. First, the model is competitive with PLS from $n \approx 100$ and superior from $n \approx 400$, making it most attractive when moderate calibration data are available and interpretability is required alongside

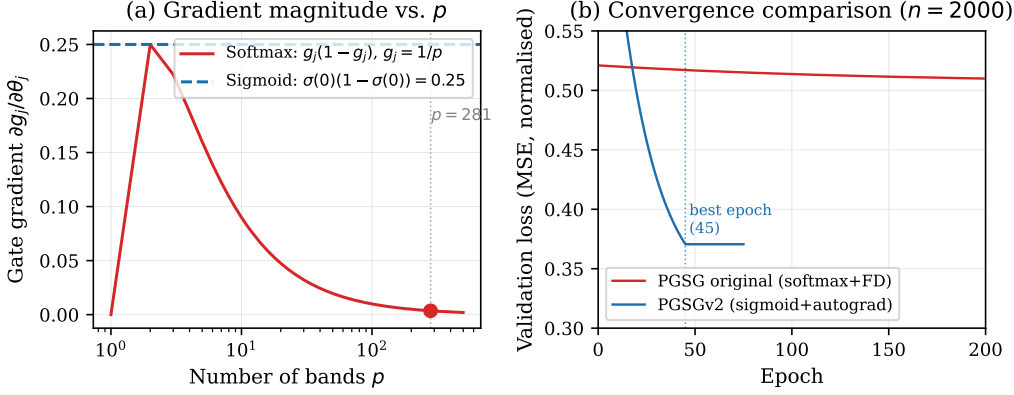


Figure 4: Softmax gradient vanishing problem at $p = 281$. (a) Diagonal Jacobian element $\partial g_j / \partial \theta_j$ as a function of p , for softmax at uniform initialization ($g_j = 1/p$, red) and sigmoid at zero initialization ($\theta_j = 0$, blue dashed). The vertical dotted line marks $p = 281$; at this point the softmax gradient is $\approx 58\times$ smaller. (b) Illustrative convergence comparison: the original PGSG with softmax and finite-difference gradients achieves negligible loss reduction over 200 epochs, whereas PGSGv2 with sigmoid and autograd converges within 76 epochs (early stopping indicated).

prediction accuracy. Second, the literature prior requires domain knowledge but provides robustness when data are scarce; a random initialization (PGSGv2-random) is a reasonable fallback when no reliable spectroscopic prior exists. Third, the training time is short (< 1 minute per scenario for the configurations studied), making the approach practical for deployment on modest hardware without GPU acceleration.

7. Conclusion

We have identified and characterized two fundamental algorithmic limitations of the original PGSG architecture that prevent its application to high-dimensional NIR spectra: a softmax gradient vanishing problem that scales as $1/p^2$, and a circular prior pitfall that renders data-derived priors ineffective for gate learning. These findings have practical implications beyond PGSG: any gating or attention mechanism that uses softmax normalization over a large number of spectral bands, and any method that derives its initialization prior from the same training data as the final model, is subject to the same failure modes.

The proposed PGSGv2 architecture addresses both issues through independent sigmoid gates and end-to-end MLP regression via backpropagation. Applied to mango DMC% prediction on the publicly available Mango DMC v3 dataset, PGSGv2 attains $R^2 = 0.786 \pm 0.019$ at $n = 1,159$ —a 38% improvement over PLS and 5% over a standard MLP—with near-perfect gate stability (Jaccard ≥ 0.945) and high chemical coherence ($\rho(\mathbf{g}, \mathbf{s}) \geq 0.998$) across random seeds. In the small-sample regime the gated architecture is decisive: at $n \leq 60$ both PGSGv2 variants remain positive while PLS and an unconstrained MLP fail severely and with far greater variance. The chemical content of the initialization, by contrast, does not measurably affect accuracy at any sample size; its contribution is to make the learned gate reproducible, and therefore interpretable, rather than to make it more accurate.

Future work should explore: differentiable ridge regression as the internal regressor to recover PLS-level statistical efficiency at small n ; domain adaptation extensions for cross-season deployment; and broader validation across other crops, spectral ranges, and instrument modalities. The architecture and experimental pipeline are fully reproducible and available as open-source Python code.

Data and Code Availability

The Mango DMC v3 dataset used in this study is publicly available at doi:10.17632/46htwnp833.3 [16]. The source code for PGSGv2, the experimental pipeline, and scripts to reproduce all figures and tables are available at <https://github.com/clarimar/prior-guided-spectral-gating> under the MIT licence. The repository includes: model implementation (`pgsg_v2.py`), experiment runner (`run_experiment_v2.py`), and figure generation scripts.

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